

Introduction to Machine Learning (ML)

ML Workflow and Real-World Applications

Daffodil International University

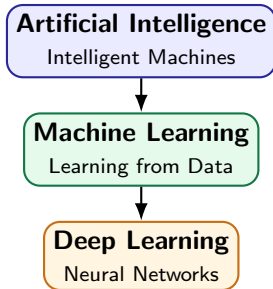
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Learning Objectives

After completing this lecture, students will be able to:

- Explain the relationship between AI, ML, and Deep Learning.
- Understand the motivation behind Machine Learning.
- Describe different types of ML.
- Explain the complete ML workflow.
- Identify real-world applications and challenges of ML.

Artificial Intelligence vs Machine Learning vs Deep Learning



Relationship

AI is the broader field.
ML learns from data.
DL uses neural networks.

Artificial Intelligence (AI)

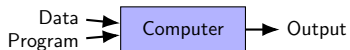
- Broad field of creating intelligent systems.
- Performs human-like tasks.
- Examples:
 - Expert Systems
 - Robotics

Machine Learning (ML)

- Subset of AI that learns patterns from data.
- Improves through experience.
- Examples:
 - Regression
 - Decision Trees

Traditional Programming vs Machine Learning

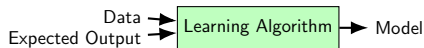
Traditional Programming



Data + Program → Output

Human defines rules manually.

Machine Learning



Data + Output → Model

Computer learns rules from examples.

Key Difference

Traditional Programming: Humans write the rules. Machine Learning: Algorithms learn the rules from data.

Components of an ML System



Data

- Raw examples collected from real-world sources.
- Foundation of every ML system.

Features

- Important characteristics extracted from data.
- Used as input for learning algorithms.

Core Idea

An ML system converts raw data into useful knowledge through features, algorithms, and trained models.

Types of Machine Learning

Supervised Learning

Data: Labeled Data (X, Y)

- Learns input-output relationships.
- Used for prediction tasks.
- Examples: Classification, Regression

Unsupervised Learning

Data: Unlabeled Data (X)

- Discovers hidden patterns.
- Finds structure in data.
- Examples: Clustering, PCA

Semi-Supervised Learning

Data: Few Labeled + Many Unlabeled Samples

- Combines supervised and unsupervised learning.
- Useful when labeling is expensive.
- Examples: Image Classification, Medical Data

Reinforcement Learning

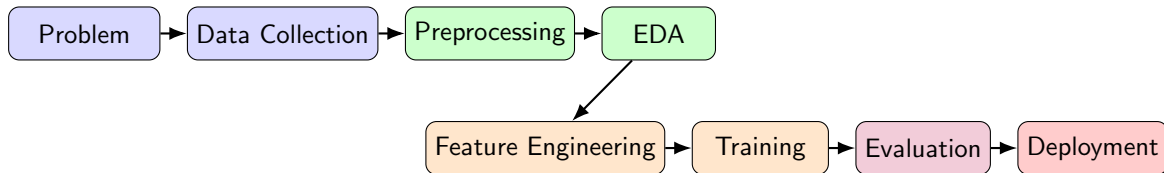
Data: States, Actions, Rewards

- Learns through interaction with environment.
- Improves using rewards and penalties.
- Examples: Robotics, Game AI

Machine Learning learns from different types of feedback:

Labels → Supervised Patterns → Unsupervised Rewards → Reinforcement

Complete ML Workflow



Data Stage

Collect data, clean missing values, and analyze patterns.

Model Stage

Engineer features, train models, and evaluate performance.

Deployment Stage

Deploy the model and monitor real-world performance.

ML Workflow Example: Student Performance Prediction

Dataset:

Study Hours	Attendance	Assignment	Result
5	90%	Good	Pass
2	50%	Poor	Fail

Features: Study Hours, Attendance, Assignment

Prediction: Pass / Fail

Common evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score
- RMSE

A good model should generalize well to unseen data.

Real-World Applications of ML

Domain	Applications
Healthcare	Diagnosis, Medical Imaging
Finance	Fraud Detection
Transportation	Autonomous Vehicles
Education	Personalized Learning
Business	Recommendation Systems
Cybersecurity	Threat Detection

Challenges in Machine Learning

- Poor quality data
- Insufficient training examples
- Overfitting
- Bias and fairness issues
- Model interpretability
- Computational requirements

Key Takeaways

- ML enables computers to learn patterns from data.
- ML requires quality data, suitable features, and proper evaluation.
- ML workflow transforms raw data into intelligent decisions.
- ML is widely used across different industries.

Thank You
Questions?